DC IA4

Elaborate on the following topics with examples:

- Sequential circuits that are not synchronized by a clock
- Analysis of asynchronous sequential circuits
- Hazards that cause incorrect behaviour of a circuit
- Synthesis of asynchronous sequential circuits
- The concept of stable and unstable states
- Hazards that cause incorrect behaviour of a circuit
- Timing issues in digital circuits

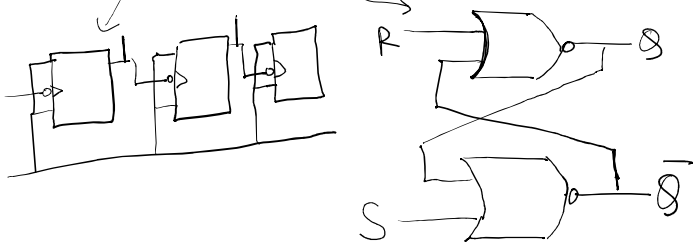
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Sequential circuits that are not synchronized by a clock

Sequential circuits are ckts whose output not only depends on current input but also on past inputs when a synchronous ckt does not use a clock signal to synchronize state changes it is called an Asynchronous Sequential ckt

Unlike synchronous sequential ckts asynchronous changes state immediately in response to input changes.

Eg → Ripple counter, SR latch



Here changes in S or R instantly affect the output → no clock is required

Comparison with synchronous circuits

Timing control	By clock pulses	By signal propagation delay
Storage elements	Flip flops (edge triggered)	Latches or feedback loops
Speed	Slower (limited by clock)	Faster
Design Complexity	Easier	More difficult (hazards & races)

Analysis of asynchronous sequential circuits

The analysis of asynchronous sequential circuits proceeds in a similar way to synchronous but with additional timing considerations.

Step 1: Identify input, Outputs and feedback paths

Step 2: Write Boolean Equations
for → Next state variables (before delay element)
→ Output variables (if any)

Step 3: Construct tables
Several forms of tables are used

Excitation table — shows req. input conditions for State change

Transition Table — Maps current input & State to next State

State Table — Lists all combinations of Present states, inputs, next States and Outputs.

Flow Table — Simplified version of State table used specifically for asynchronous circuits, showing stable and unstable states.

Step 4 → Determine Stable States → a condition where ckt no longer changes until input changes

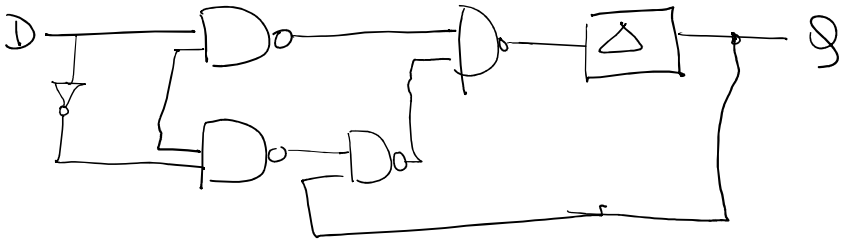
Step 4 → Determine Stable States → a condition where ckt no longer changes until input changes

Step 5 → Draw flow table & State diagram

Examples → gated D latch

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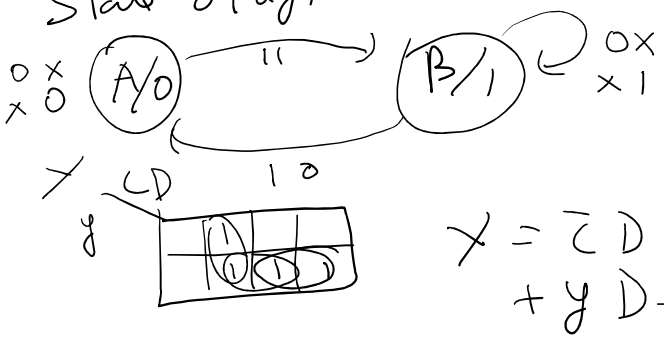
State table

PS	NS(Y) CD 00	01	10	11	Output Q
y					
0	0	0	0	1	0
1	1	1	0	1	1

Flow table

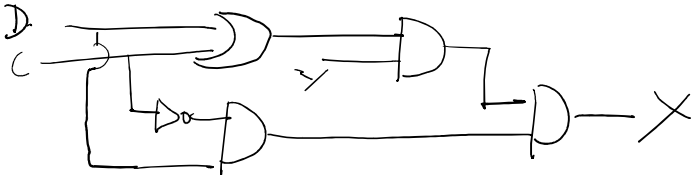
PS	NS CD 00	01	10	11	Output Q
A	A	A	A	B	0
B	B	B	A	B	1

State diagram



$$Y = \bar{C}D + yD + yC + y(D+C) + \bar{C}D$$

We use additional gate to eliminate hazard



Hazards that cause incorrect behaviour of a circuit

Hazards are unwanted transient changes in output caused by unequal propagation delays through logic gates.

Types of hazards:

→ Static hazards: Output should remain constant but momentarily changes

→ Dynamic hazards: Output changes multiple times instead of once during transitions

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Example: For ckt $F = A'B + AB'$

If A & B simultaneously change from (0,1) to (1,0) a glitch may occur due to different gate delays.

This could cause the output to temporarily drop to 0 before stabilizing at 1.

Hazards can cause unintended state transitions in asynchronous circuits since they rely on immediate input changes.

Synthesis of asynchronous sequential circuits

① Devise a state diagram
Create a state diagram for FSM that realizes the req'd. behavior

② Derive & simplify the flow table
→ Translate the state diagram into a flow table
→ Reduce the number of states if possible by merging equivalent ones.

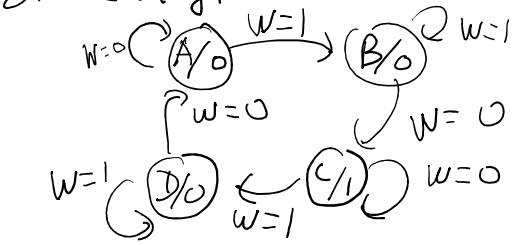
③ Perform state assignment and derive the excitation table
→ Assign binary codes to the states
→ Derive the excitation table that shows latch input requirements for state transitions

④ Obtain next state & output expressions
→ Derive Boolean equations for next state & output logic

⑤ Construct the ckt
→ Implement the expressions

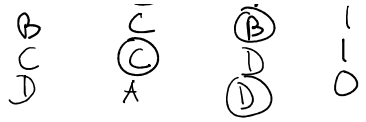
Eg → Serial parity generator

State diagram



Flow table

PS	NS		Output
	W=0	W=1	Z
A	(A)	B	0
B	C	(B)	1
C	(C)	D	1
D	A	(D)	0



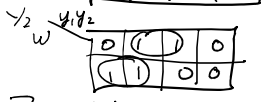
State Assignment

PS	NS w=0	NS w=1	output z
$y_1 y_2$	$Y_1 Y_2$	$Y_1 Y_2$	
00	00	01	0
01	11	01	1
11	11	10	1
10	00	10	0

Kmap



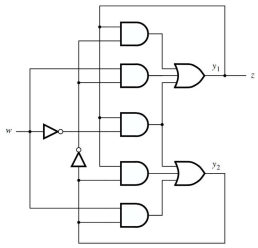
$Y_1 = \bar{w} y_2 + w y_1 + y_1 y_2$



$Y_2 = \bar{w} y_2 + w \bar{y}_1 + \bar{y}_1 y_2$



$z = y_2$



The concept of stable and unstable states

In asynchronous ckts, a stable state is one where ckt will remain indefinitely until an input changes.
An unstable state is one where internal feedback to transition automatically to another state, even without input change

Eg → In an SR latch
→ if S=0, R=0 → ckt remains stable (holds its prev state)
→ if S=R=1 → both outputs go to 0 (invalid, unstable cond) because feedback tries to drive contradictory outputs
Stable states corresponds to point where NS = PS

Timing issues in digital circuits

Timing issues arise due to:

→ Gate propogation delays
→ Set up & hold time violations
→ Clock skew or jitter (in synchronous systems)
→ Race conditions (in asynchronous systems)

Eg → If two signals arrive at a ff slightly out of sync one may violate the setup time causing unpredictable output (metastability)

In asynchronous ckts main timing issue is race conditions.

A race condition occurs when a single input change requires two or more state variables to change at the same time

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Eg →

Present state y_2y_1	Next state $w = 0$	Next state $w = 1$	Output z
	Y_2Y_1		
00	(00)	01	0
01	10	(01)	1
10	(10)	11	1
11	00	(11)	0

(a) Poor state assignment

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